Cpts 317. 12/2

Last time: Halting problem for TMs is undecidable; there is no alg to decide whether a C-program is going to halt or not.

Real-word undecidable problem: PCP.
(Post Corresp. Problem)

Col 1	Col 2
α_1	β_1
α_2	β_2
$\vdots$	$\vdots$
α_k	β_k

Want: a seq of numbers, such as

2 1 1 2 2 3 8

From col 1,

$$\alpha_2 \alpha_1 \alpha_1 \alpha_2 \alpha_2 \alpha_3 \alpha_8 = \alpha$$

From col 2,

$$\beta_2 \beta_1 \beta_1 \beta_2 \beta_2 \beta_3 \beta_8 = \beta.$$

to make $\alpha == \beta$.

Gödel's Thm: there is a true statement in Math that can't be proven in Math.

$$\text{Math} \vdash t$$

prove

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proofs are carried out by using logic-rules & Axioms.

$$\frac{A \quad , \quad A \to B}{B}$$

} modus ponens.

$\underbrace{\text{Math} \vdash}$ t

Can be implemented
by a TM or a C-program
That does string rewriting.

equivalent to a TM M running on input string t s.t.
M says yes on t iff Math $\vdash t$.

For each statement t, either t or $\bar{t}$ is true.

Suppose that Math can prove every true statement.

Then, we run

M on t and

M on $\bar{t}$, in parallel.

Then, either M says yes on t or

M says yes on $\bar{t}$, for each t.

Let $\mathcal{M}$ and w be a TM and a word.

"$\mathcal{M}$ halts on w" is a statement σ.

Then, M says yes on σ or

M says yes on $\overline{\sigma}$.

$\Rightarrow$ "$\mathcal{M}$ halts on w" is decidable. $\times$

Examples: Math H

"Math doesn't have contradiction"

$$\left.\begin{array}{c} N \prec \aleph \prec R \\ \uparrow \\ \text{Cantor's} \end{array}\right\} \text{CH.}$$

(Continum Hypothesis)

Human's logic reasoning is Turing.

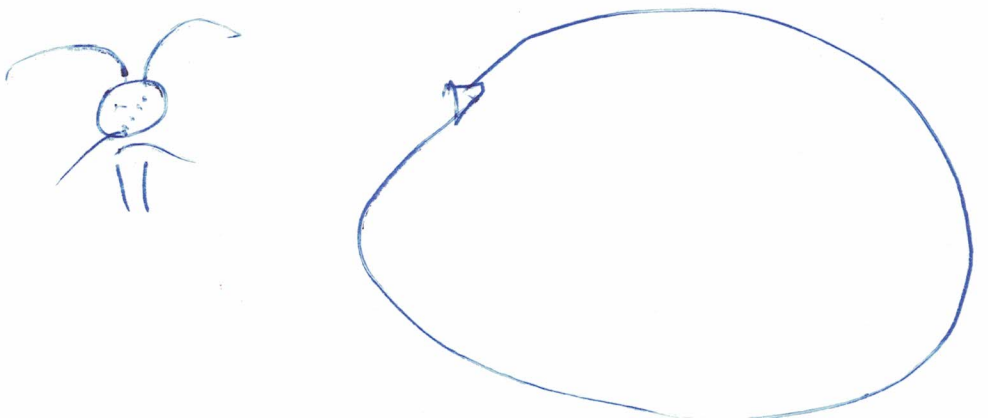
Our logic reasoning is from ?

3000 yrs ago.

$$\frac{A \quad , \quad A \to B}{B} \quad ,$$

why? ← why all humans share the same logic system ?

$$\frac{A \to B}{}$$



movie: Arrival.